

EUTECH ENGINEERS

REGULATED 5V POWER SUPPLY

L7805CV Linear Regulator

DIENG Assane

12/03/2026

TABLE OF CONTENTS

1	Introduction.....	2
2	Specifications	3
3	Overview of the L7805CV Regulator	3
4	Electronic Schematic.....	4
5	Component sizing	5
5.1	Input filter capacitor	5
5.2	High-frequency bypass capacitor	6
5.3	Output filter capacitor	6
5.4	LED limiting resistor.....	7
5.5	Thermal calculation for regulator U1	7
5.5.1	Junction temperature with SK129 heat sink.....	7
5.6	Component summary table.....	8
6	LTspice simulation	8
6.1	Load response.....	9
7	Conclusion	10
8	References.....	11
9	Appendix.....	11

1 Introduction

This report presents the work carried out as part of Task 1 of the internship at STMicroelectronics. The objective is the complete design of a 5V regulated power supply using the L7805CV linear regulator, from the circuit diagram through PCB routing to SPICE simulation.

The deliverable includes:

- A complete electronic schematic created in KiCad
- A routed and verified single-layer PCB
- Gerber files ready for manufacturing
- LTSpice simulations validating operation
- A report
- All versioned on GitHub (Assane401)

2 Specifications

Parameter	Value	Note
Input voltage V_{in}	7V to 35V	Recommended: 8V to 12V
Output voltage V_{out}	$5.0V \pm 2\%$	Regulated, fixed
Max. output current	1.5A absolute	500mA at rated load
Ripple frequency	100 Hz	Full-wave rectification
Thermal protection	Integrated L7805CV	$T_{j_max} = 125^{\circ}C$
Visual indicator	Green LED	Power supply present
PCB layer	Single-layer	Copper only
PCB dimensions	125mm $\times$ 45mm	Compact format

3 Overview of the L7805CV Regulator

The L7805CV is a 5V fixed positive voltage regulator from STMicroelectronics' L78xx family. It natively integrates current limiting, thermal protection, and short-circuit protection, making it particularly robust for industrial applications [1].

Parameter	Value	Condition
Output voltage	$5.0V \pm 2\%$	$V_{in} = 10V$, $I_{out} = 500mA$
Max. output current	1.5A	With adequate heat sink
Min. input voltage	7V (2V dropout)	$V_{out} + V_{dropout}$

Max input voltage	35V	Absolute
Max. junction temperature	125°C	Built-in thermal protection
Thermal resistance Rth_jc	5°C/W	TO-220
Package	TO-220-3	Pin1=VI, Pin2=GND, Pin3=VO

4 Electronic Schematic

The schematic follows the typical application circuit recommended by the STMicroelectronics datasheet [1], with the addition of an LED indicator circuit.

Ref.	Component	Value	Function
J1	2-pin header	2.54 mm	Input connector (Vin / GND)
C1	Electrolytic	100μF / 25V	Low-frequency input filter (100 Hz)
C2	X7R ceramic	330nF / 50V	High-frequency bypass (HF stability)
U1	L7805CV	5V / 1.5A	Linear voltage regulator
C3	Electrolytic	10μF / 16V	Output filtering and transient stability
R1	1/4W resistor	150Ω	LED current limiting
D1	Green LED 3mm	Vf = 2V	Power indicator on
J2	2-pin header	2.54 mm	Output connector (Vout / GND)

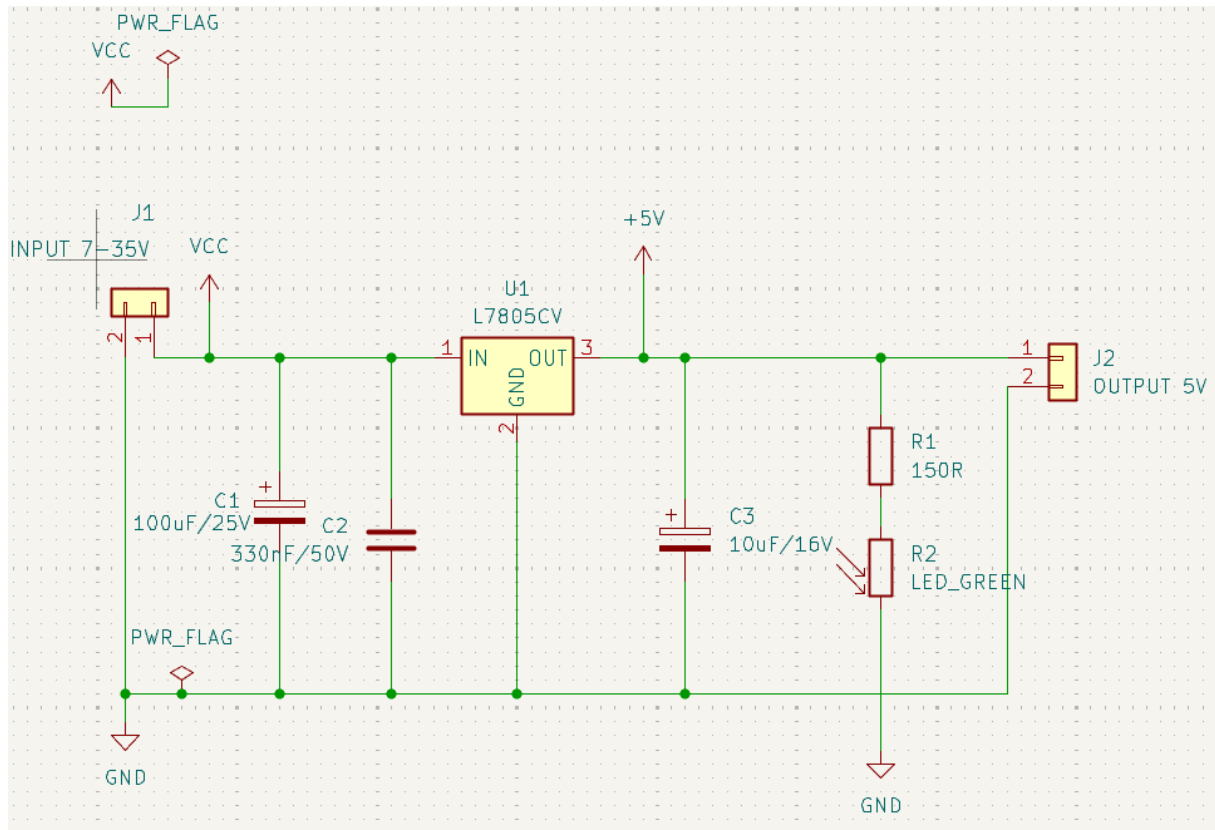


Figure 1: Regulator schematic

5 Component sizing

5.1 Input filter capacitor

C1 filters low-frequency ripple (100 Hz) at the regulator input and stores the energy required between two voltage peaks.

$$C_1 = \frac{I_{out}}{2f\Delta V} = 2500 \mu F$$

Where:

- I_{out} = 0.5 A maximum load current
- f = 100 Hz mains frequency after rectification
- ΔV = 1 V allowable ripple at the input

The L7805CV datasheet recommends $C1 \geq 0.33 \mu F$. The $100 \mu F$ value improves ripple filtering and protects against transient voltage drops.

Higher commercial value selected for safety margin: $C1 = 100 \mu F / 25V$ (minimum recommended datasheet value [1])

5.2 High-frequency bypass capacitor

C2 filters high-frequency disturbances (noise, switching) on the power rail. It is placed directly between IN and GND, as close as possible to the regulator.

$$f_c = \frac{1}{2\pi RC_2} \rightarrow C_2 = 15.9 nF$$

Objective: attenuate noise starting at 10 kHz. And $R=1 \Omega$

The L7805CV datasheet requires $C2 \geq 100 nF$ at the input for stability. A value of 330 nF is recommended to eliminate HF oscillations.

Commercial value chosen higher for margin: $C2 = 330 nF / 50V$

5.3 Output filter capacitor

C3 improves the stability of the output regulator and filters residual ripple. It determines the transient response during load steps.

$$\Delta V_{out} = \frac{I_{out}\Delta t}{C_3} \rightarrow C_3 = 10 \mu F$$

Where:

- $I_{out}=0.5 A$ step current
- $\Delta t= 1 \mu s$ controller response time
- $\Delta V_{out} \leq 50 mV$ allowable voltage drop

5.4 LED limiting resistor

R1 limits the current flowing through the LED D1 to protect it from destruction by overcurrent, while ensuring satisfactory illumination.

$$R1 = \frac{V_{out} - v_f}{I_f} \rightarrow R1 = 150 \Omega$$

Where:

- V_{out} = 5 V regulated output voltage
- v_f = 2 V LED forward voltage
- I_f = 20 mA LED rated current

5.5 Thermal calculation for regulator U1

Power dissipation

$$P_d = (V_{in} - V_{out})I_{out}$$

$$P_d = (12 - 5) * 0.5 = 3.5 W$$

Junction temperature without heat sink

$$T_j = T_a + R_{th_ja} * P_d$$

With $R_{th_ja} = 50^\circ\text{C/W}$ (TO-220 without heat sink): $T_j = 200^\circ\text{C}$

We have noted that $T_j = 200^\circ\text{C} > T_{j_max} = 125^\circ\text{C}$, so a heat sink is required.

5.5.1 Junction temperature with SK129 heat sink

$$T_j = T_a + (R_{th_jc} + R_{th_hs}) * P_d$$

Where:

- $R_{th_jc} = 5^\circ\text{C/W}$, junction-housing L7805CV
- $R_{th_hs} = 3.8^\circ\text{C/W}$, Fischer SK129 heat sink

$$T_j = 25 + (5 + 3.8) \times 3.5$$

$$T_j = 25 + 8.8 \times 3.5 = 25 + 30.8 = 55.8^\circ\text{C}$$

$T_j = 55.8^\circ\text{C} < 125^\circ\text{C} \rightarrow$ Safe operation with SK129

5.6 Component summary table

Ref.	Value	Formula	Calculation result	Status
C1	100 μF	$C_1 = \frac{I_{out}}{2f\Delta V}$	2,500 μF min $\rightarrow$ 100 μF datasheet	Recommended datasheet
C2	330 nF	$f_c = \frac{1}{2\pi RC_2}$	15.9 nF min $\rightarrow$ 330 nF selected	HF stability ensured
C3	10 μF	$\Delta V_{out} = \frac{I_{out}\Delta t}{C_3}$	10 μF for $\Delta V \leq 50$ mV	Simulation: 110 mV
R1	150 Ω	$R1 = \frac{V_{out} - v_f}{I_f}$	150 Ω	$P = 60$ mW < 250 mW
D1	Green LED	$V_f = 2\text{V}$, $I_f = 20\text{mA}$	20 mA	Nominal lighting
U1	L7805CV	$T_j = T_a + (R_{th_{jc}} + R_{th_{hs}}) * P_d$	$T_j = 55.8^\circ\text{C}$ with SK129	$< 125^\circ\text{C}$ — Heatsink required

6 LTspice simulation

Since the L7805CV does not have an official SPICE model provided by STMicroelectronics, the LT1085-5 model from Analog Devices was used as a functional equivalent. The electrical characteristics are very similar ($V_{out} =$ fixed 5V, same family of linear regulators), with a slightly better dropout voltage (1.5V vs. 2V) but identical behavior at the operating frequency.

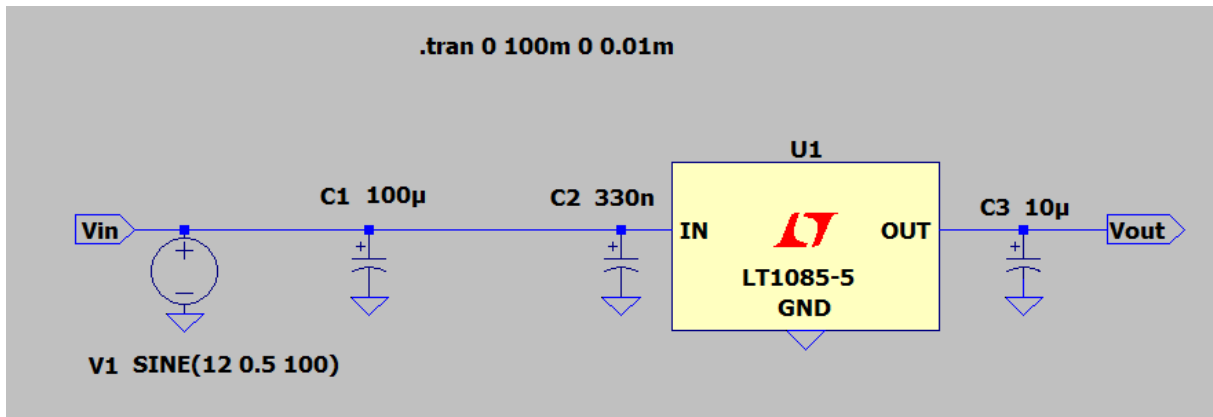


Figure 2: LTspice schematic

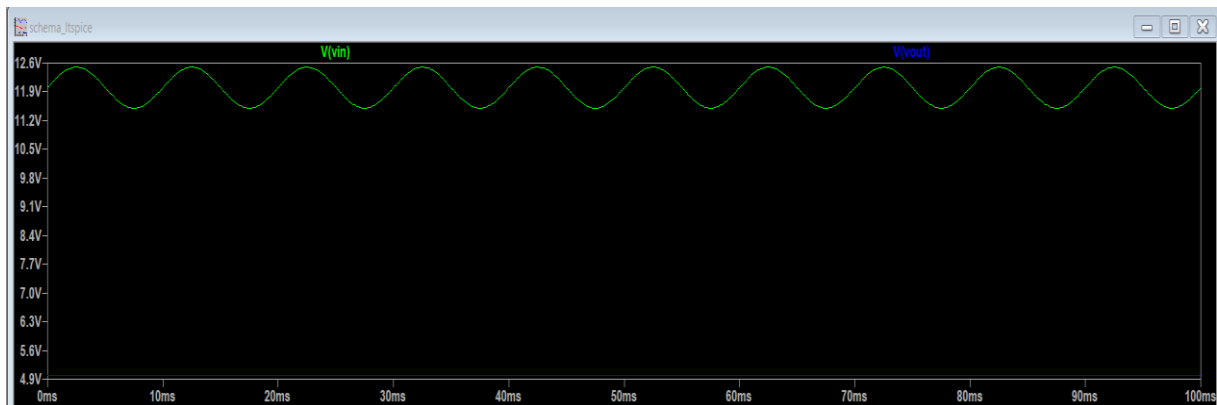


Figure 3: Simulation results without load

The simulation confirms that the regulator operates correctly; regardless of the input voltage value between 7 and 35 V, the output voltage is always regulated to 5 volts.

6.1 Load response

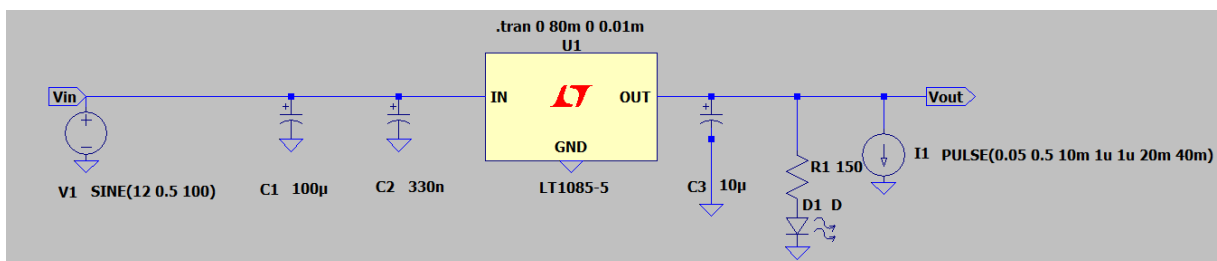


Figure 4: Schematic with load in LTspice

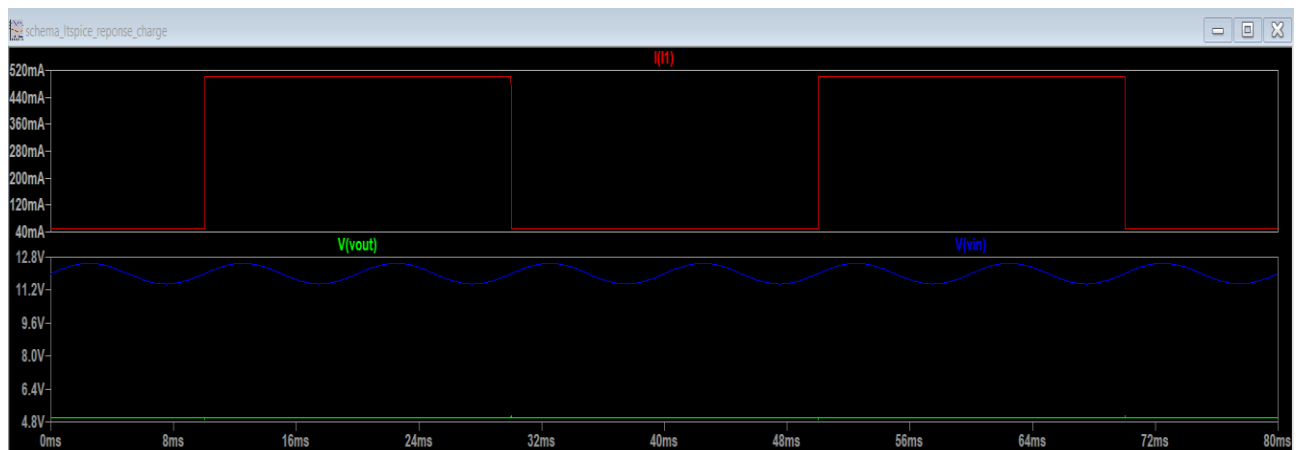


Figure 5: Simulation results with load

The regulator is perfectly stable; the output voltage is 5 V, despite a sudden load step from 50 mA to 480 mA and an input ripple of 3.2 V.

7 Conclusion

This Task 1 enabled the complete design of a 5V regulated power supply based on the STMicroelectronics L7805CV regulator. All stages electronic schematic, PCB routing, SPICE simulation, and component justification were successfully completed.

The LTSpice simulations confirm the validity of the design: the output voltage is regulated to 5.003 V ($\pm 0.006\%$), the residual ripple is less than 0.3 mV, and the load response also shows good dynamics of the regulated voltage.

Thermal analysis shows that a Fischer SK129 heat sink is essential to maintain the junction temperature at 55.8°C under a 500mA load with $V_{in} = 12V$, well below the absolute limit of 125°C.

All files are available on GitHub (Assane401) under the tag v1.0-submit, ready for review and production.

8 References

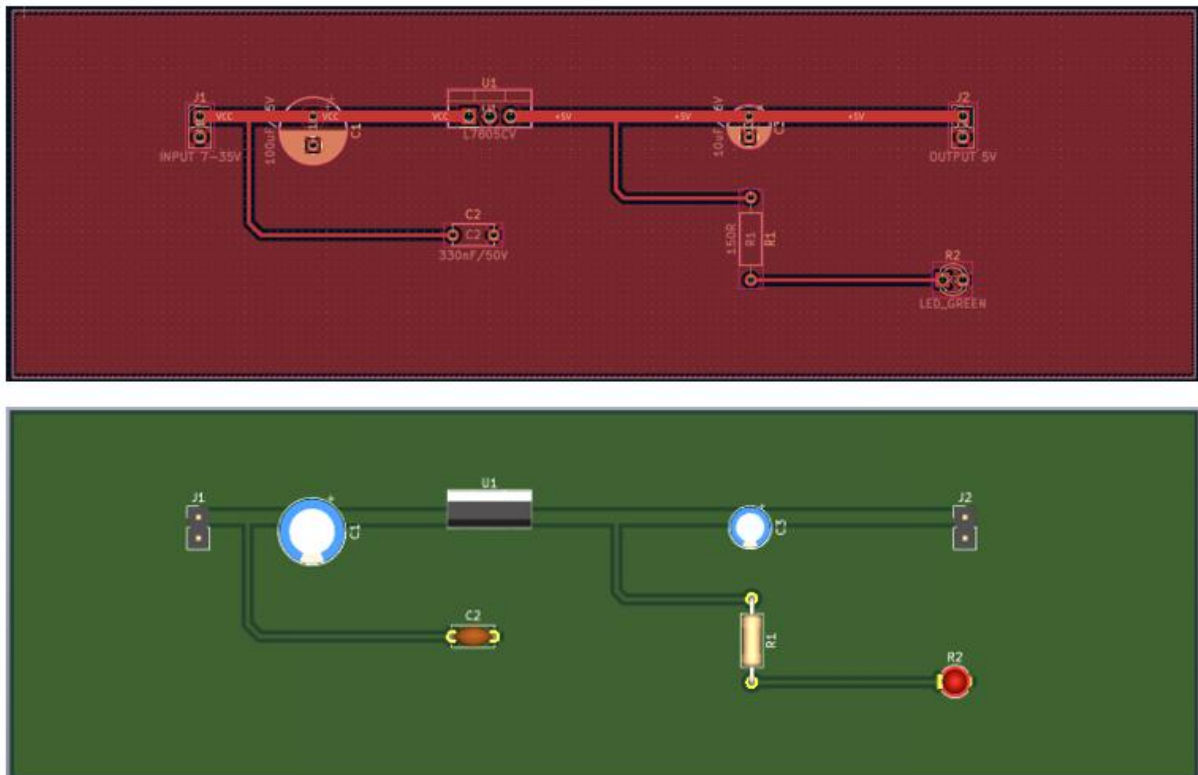
[1] STMicroelectronics “Official L7805CV Datasheet.” Online:

<https://www.st.com/resource/en/datasheet/l78.pdf>

LIST OF FIGURES

Figure 1: Regulator schematic.....	5
Figure 2: LTspice schematic	9
Figure 3: Simulation results without load	9
Figure 4: Schematic with load in LTspice	9
Figure 5: Simulation results with load	10

9 Appendix



Appendix 1: PCB in KiCad